

A model of macroalgal decomposition

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Tables

Table S1. Model diagnostics for comparison of $k(t)$ parameterisations. Convergence scores ($\hat{R}$) and effective sample sizes (ESS) for 10^3 iterations $\times$ 8 chains are given as mean $\pm$ standard deviation. Expected log predictive density (ELPD), a measure of predictive accuracy, ELPD relative to the best performing model per study (Δ ELPD), and the difference between ELPD and the log predictive density (LPD – ELPD), a measure of overfitting, are given as mean $\pm$ standard error.

Parameterisation	$\hat{R}$	ESS	ELPD	Δ ELPD	LPD – ELPD
Frontier et al. 2022					
Constant intercept	1 $\pm$ 0.00039	3481 $\pm$ 1682	50 $\pm$ 16	0 $\pm$ 0	5.1 $\pm$ 1.4
Constant slope	1.01 $\pm$ 0.0026	756 $\pm$ 145	46 $\pm$ 16	-3.9 $\pm$ 1.3	8.1 $\pm$ 2
No constants	1.01 $\pm$ 0.002	1309 $\pm$ 404	47 $\pm$ 16	-3 $\pm$ 0.97	7.2 $\pm$ 1.7
Vandendriessche et al. 2007					
Constant intercept	1 $\pm$ 0.0049	6994 $\pm$ 2879	1132 $\pm$ 102	-280 $\pm$ 74	110 $\pm$ 28
Constant slope	1 $\pm$ 0.0085	6680 $\pm$ 3046	1402 $\pm$ 101	-14 $\pm$ 9.7	127 $\pm$ 27
No constants	1.02 $\pm$ 0.096	7887 $\pm$ 3932	1416 $\pm$ 102	0 $\pm$ 0	126 $\pm$ 25

Table S2. Model diagnostics for comparison of likelihood functions. See Table S1 for definitions.

Likelihood	$\hat{R}$	ESS	ELPD	Δ ELPD	LPD – ELPD
Frontier et al. 2022					
Beta'	1 $\pm$ 0.0012	3579 $\pm$ 1323	-130 $\pm$ 81	-180 $\pm$ 72	48 $\pm$ 46
Gamma	1 $\pm$ 0.00049	3683 $\pm$ 1501	-74 $\pm$ 65	-120 $\pm$ 56	34 $\pm$ 31
Lognormal	1 $\pm$ 0.001	3213 $\pm$ 557	-260 $\pm$ 114	-310 $\pm$ 105	53 $\pm$ 52
Normal	1 $\pm$ 0.0016	3383 $\pm$ 1478	50 $\pm$ 16	0 $\pm$ 0	5.2 $\pm$ 1.4
Vandendriessche et al. 2007					
Beta'	1 $\pm$ 0.0046	7306 $\pm$ 3202	1282 $\pm$ 178	-32 $\pm$ 34	153 $\pm$ 47
Gamma	1 $\pm$ 0.0049	9929 $\pm$ 3728	1314 $\pm$ 151	0 $\pm$ 0	137 $\pm$ 38
Lognormal	1 $\pm$ 0.0027	9855 $\pm$ 4327	625 $\pm$ 246	-690 $\pm$ 251	245 $\pm$ 98
Normal	1 $\pm$ 0.004	6639 $\pm$ 2939	1132 $\pm$ 102	-180 $\pm$ 90	112 $\pm$ 29

Table S3. Model diagnostics for homo- vs. heteroskedasticity. See Table S1 for definitions.

Variance	$\hat{R}$	ESS	ELPD	Δ ELPD	LPD – ELPD
Frontier <i>et al.</i> 2022					
Beta' likelihood					
Heteroskedastic	1 ± 0.0013	4188 ± 1526	31 ± 38	-20 ± 6.8	23 ± 19
Homoskedastic	1 ± 0.0016	3359 ± 1167	-130 ± 81	-180 ± 54	45 ± 43
Gamma likelihood					
Heteroskedastic	1 ± 0.00081	3842 ± 1556	51 ± 31	0 ± 0	22 ± 18
Homoskedastic	1 ± 0.00043	3537 ± 1481	-76 ± 68	-130 ± 41	35 ± 33
Vandendriessche <i>et al.</i> 2007					
Beta' likelihood					
Heteroskedastic	1 ± 0.02	8837 ± 3805	1320 ± 138	-66 ± 36	169 ± 49
Homoskedastic	1 ± 0.004	7324 ± 3162	1284 ± 178	-100 ± 78	148 ± 44
Gamma likelihood					
Heteroskedastic	1 ± 0.016	7455 ± 3509	1385 ± 105	0 ± 0	148 ± 34
Homoskedastic	1 ± 0.0092	9124 ± 3389	1319 ± 146	-66 ± 47	133 ± 35

Table S4. Model diagnostics for macroalgal vs. conventional model. See Table S1 for definitions.

Model	$\hat{R}$	ESS	ELPD	Δ ELPD	LPD – ELPD
Frontier <i>et al.</i> 2022					
Macroalgal	1 ± 0.00045	4248 ± 1421	31 ± 37	-15 ± 28	23 ± 19
Conventional	1 ± 0.00075	4557 ± 1337	46 ± 16	0 ± 0	3.9 ± 1.4
Vandendriessche <i>et al.</i> 2007					
Macroalgal	1 ± 0.022	9007 ± 4030	1321 ± 136	0 ± 0	167 ± 46
Conventional	1 ± 0.00061	10491 ± 1187	-1000 ± 72	-2300 ± 147	13 ± 1.2

Figures

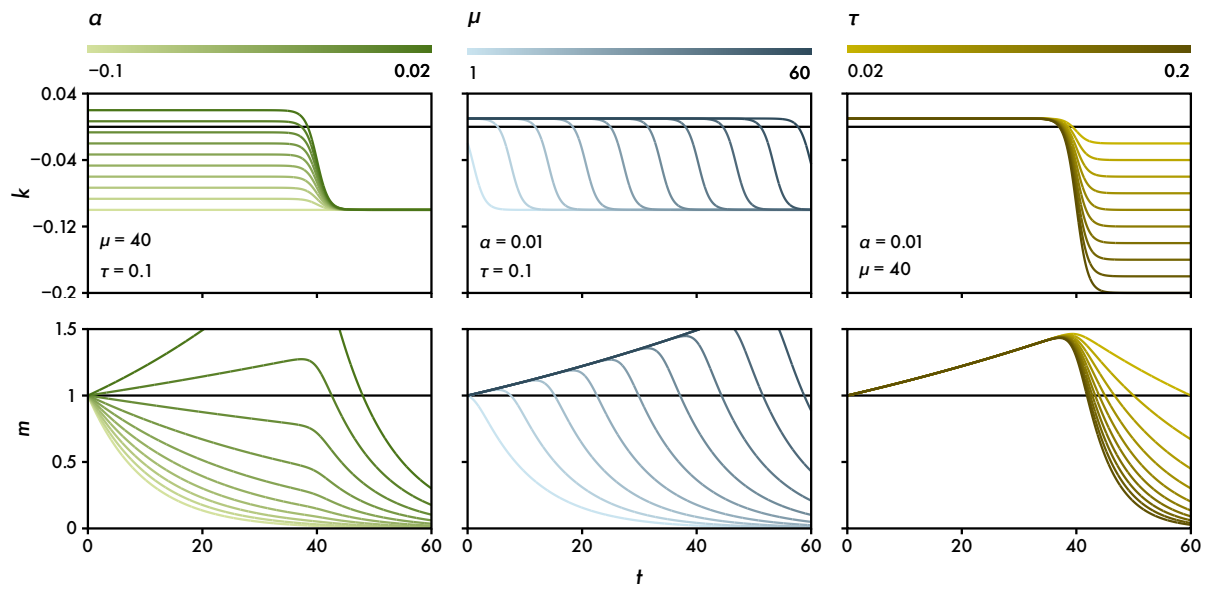


Figure S1. Constant slope parameterization of $k(t)$ (cf. Figure 1).

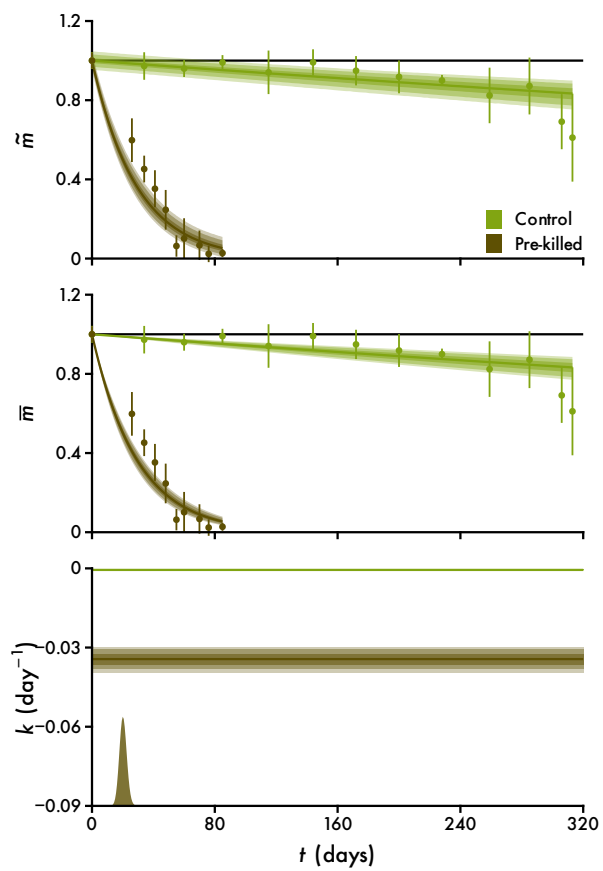


Figure S2. Conventional version of Figure 2. The distribution shows the exponential half-life $t_{1/2}$ instead of the photosynthetic half-life μ .

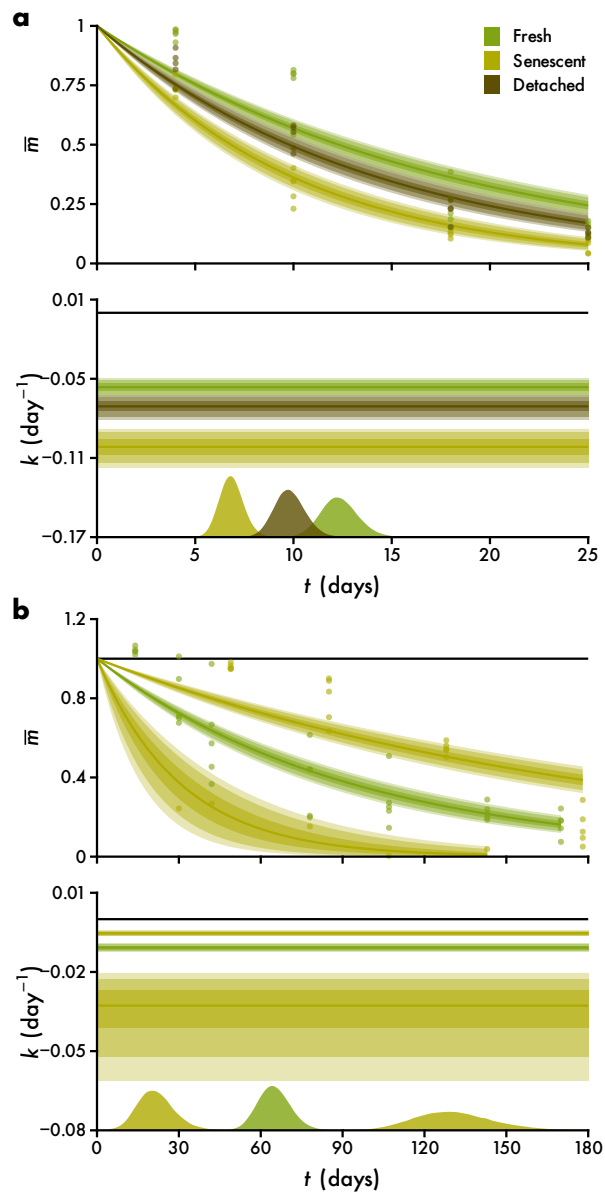


Figure S3. Conventional version of Figure 3. Distributions show the exponential half-life $t_{1/2}$ instead of the photosynthetic half-life μ .

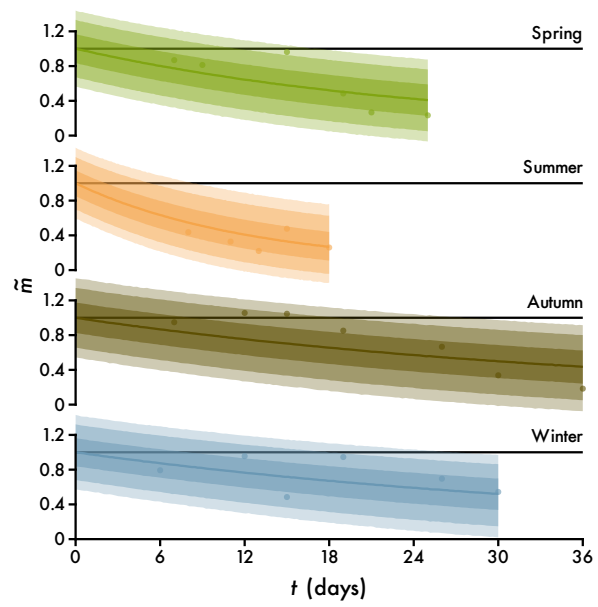


Figure S4. Conventional version of Figure 4.

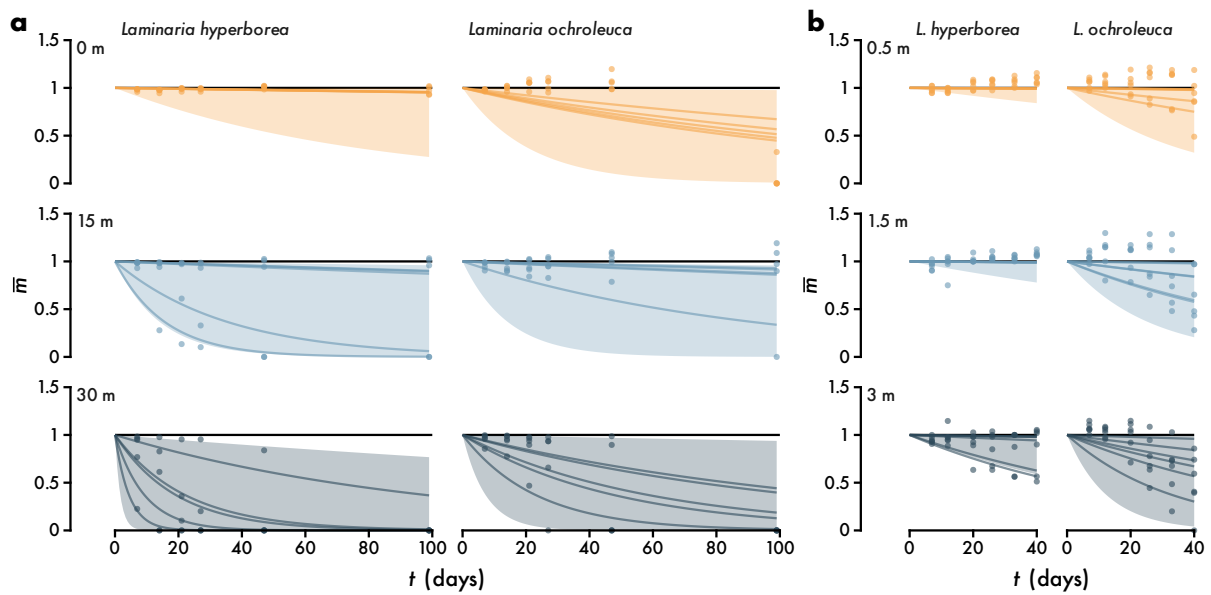


Figure S5. Conventional version of Figure 5.

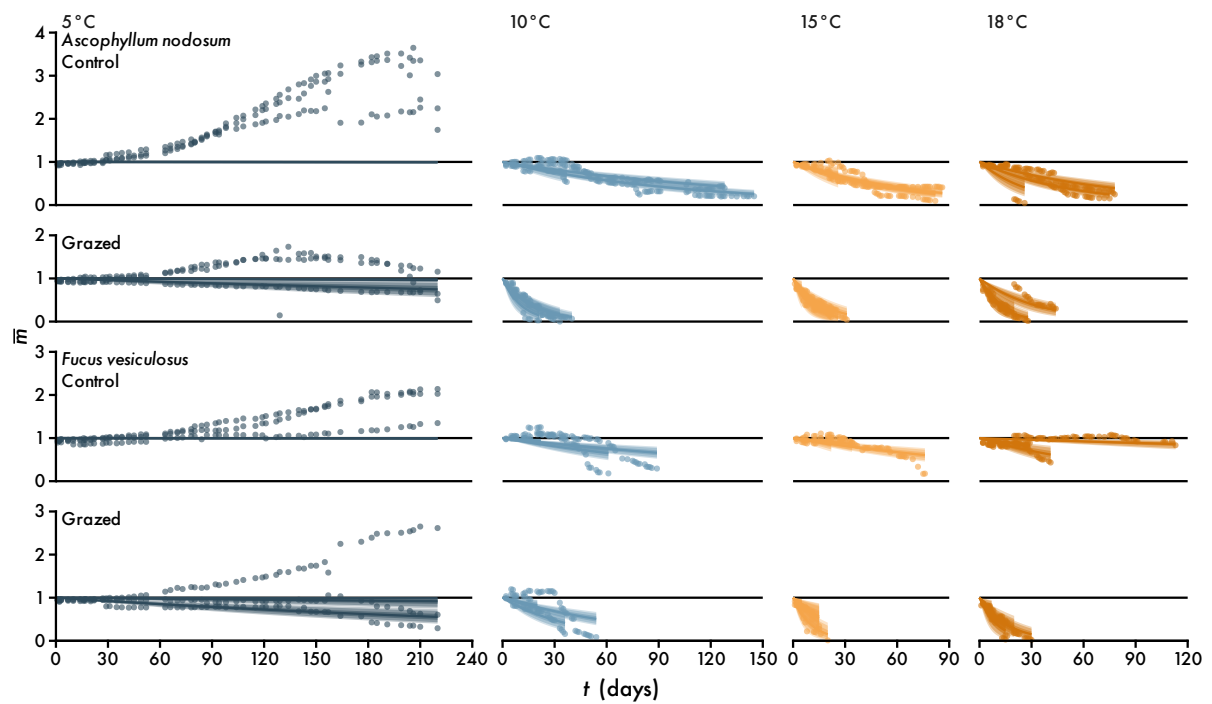


Figure S6. Conventional version of Figure 6. NB where it seems that the prediction is missing, it is masked by the horizontal line marking $m = 1$.